

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q21-1870

Sunnyslope 100 kV
February 01, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1870. The Interconnection Customer proposes a 177.5 MW solar photovoltaic facility interconnecting at the Sunnyslope 100 kV substation in the Ohio area. The request was received on August 27, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 580 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 9 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$14,589,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Sunnyslope to Ponchatoula 100 KV line	Carencro to Sunnyslope 100 KV line	177.4	195	100.0
Delta York Ec to Devon 100 KV	Pnchatoula to Sunnyslope 100 KV	160.0	181	100.0
Carecro to Devon 100 KV line	Sunnyslope to Ponchatoula 100 KV	148.8	218	100.0
Sunnyslope to Carencro Substation 100 KV	Ponchatoula to Sunnyslope 100 KV L	143.6	240	100.0
Ponchatoula to Ft Nelson 100 KV	Sunnyslope Substation to Carencro 100 KV	134.6	121	37.6
Huson to Ft Nelson 100 KV	Sunnyslope Substation to Carencro 100 KV	128.9	116	37.6
Delta York Ec to Channel 100 KV	Ponchatoula to Sunnyslope Substation 100 KV	122.8	129	41.1
Colegio 6006 to Huson 100 KV	Sunnyslope to Carencro 100 KV	113.9	114	37.6
Oglethorpe to Colegio 6006 100 KV	Carencro to Sunyslope 100 KV	101.7	119	37.6

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Sunnyslope to Ponchatoula 100 KV	Rebuild line with higher-capacity conductor	\$1,665,000	48% (\$807,000)
Delta York Ec to Devon 100 KV	Rebuild line with higher-capacity conductor	\$1,791,000	48% (\$868,000)
Carecro to Devon 100 KV line	Rebuild line with higher-capacity conductor	\$2,859,000	48% (\$1,387,000)
Sunnyslope to Carencro 100 KV	Rebuild line with higher-capacity conductor	\$1,428,000	48% (\$693,000)
Ponhatoula to Fort Nelson 100 KV L	Rebuild line with higher-capacity conductor	\$4,392,000	48% (\$2,130,000)
Huson to Ft Nelson 100 KV	Rebuild line with higher-capacity conductor	\$5,758,000	48% (\$2,792,000)
Delta York Ec to Chnnel 100 KV	Rebuild line with higher-capacity conductor	\$2,612,000	45% (\$1,169,000)
Huson to Colegio 6006 100 KV	Rebuild line with higher-capacity conductor	\$8,360,000	48% (\$4,054,000)
Oglethorpe to Colegio 6006 100 KV	Reconductor line	\$1,422,000	48% (\$689,000)
	Total allocated to project		\$14,589,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.